University of Central Punjab

Faculty of Information Technology and Computer Science

Course Title: Human Computer Interaction
Course Code: SESE 2123

Assignment 2

Date: 08-11-2025

Total Marks 50 (Assignment) +10 (Evaluation)=60

Name	Registration Number	Section

CLO #	Course Learning Outcome (CLO)	Taxonomy Level	Mapping to PLO
CLO 2	Apply user centric methods to understand and analyze population dynamics, their wants, needs and irritations.	C3	PLO 3

Instructions:

1. Attempt all questions.
2. Group Member Allowed (Maximum 2)
3. Assignment should be handwritten and submitted on A4 sheets or Assignment sheets only. Violation will result to deduction of 1 mark from the scored marks.
4. Each student will have attached this front page with his/her assignment. Violation will result to deduction of 1 mark from the scored marks.
5. Due Date for Assignment is November 14, 2025 10:00 PM (Portal).
6. Late submission will result in 10% deduction in marks.
7. No request for late submissions will be considered after two working days of the deadline.
8. No request to review assignment will be considered after 3 working days of the review in class.

Read the scenario given below and answer the questions 1-7.

Over the past few years, there has been a noticeable rise in health concerns such as obesity, anxiety, and poor sleep quality, especially among young professionals and university students. Many people subscribe to gym memberships or fitness programs but lose motivation within a few weeks because they fail to track their progress or maintain consistency.

Currently available fitness apps offer generic workout plans and calorie tracking, but users report several challenges:

- **Lack of personalization:** Most apps recommend one-size-fits-all routines without considering differences in users' physical abilities, goals, or medical conditions.
- **Overwhelming data:** Users find it hard to interpret charts and statistics without guidance. They want meaningful feedback like "You're improving your stamina by 15%" instead of just raw numbers.
- **Motivational fatigue:** After initial enthusiasm, users lose interest when there's no human-like encouragement or reminders tailored to their habits.
- **Disconnected lifestyle support:** Current apps rarely integrate with diet, sleep, and stress levels simultaneously — making it hard for users to understand how these factors influence each other.
- **Accessibility and usability barriers:** Some middle-aged users or beginners find the app interfaces cluttered and difficult to navigate. They need a simple way to track progress and receive help when confused.
- **Privacy concerns:** People hesitate to share fitness data without assurance of security and transparency.

A team of software engineering students has decided to design an AI-powered fitness companion app that addresses these problems. The app aims to integrate wearable sensors, diet tracking, and AI-based motivation. However, before design begins, the team must clearly define who the users are, what their goals and pain points are, and what problems the system must solve first.

Q1. Brainstorming (10 marks)

a) **Show** how brainstorming could help the design team in identifying innovative solutions for the AI fitness app.

b) **Predict** five creative ideas that could emerge from a brainstorming session to make the app engaging and different from existing solutions.

Q2. Target Market (5 marks)

Predict the target market for the AI fitness app using the requirements given in the scenario above.

Q3. User Classes (5 marks)

a) From the given scenario, **Predict** at least three user classes. For each class, **discover** whether its a primary or a secondary class.

b) For each class, **discover** their main fitness goal, level of technology familiarity and kind of support they would expect from the app.

Q4. Storyboard (5 marks)

a) **Use** an eight-up storyboarding technique to show a user's needs and irritations that are the main cause of creating this fitness app.

Q5. Need Finding (10 marks)

a) If you want to gather some requirements for the ideas generated in Q1, which need finding technique will you **apply**?

b) How interviews help in identifying user needs. **Apply** interview script writing to produce five interview questions related to AI fitness app.

Q6. Class Profile (5 marks)

a) **Prepare** a class profile for one primary and one secondary user class identified in Q3.

Q7. Persona (10 marks)

a) **Prepare** a persona for one potential user of the AI fitness app.